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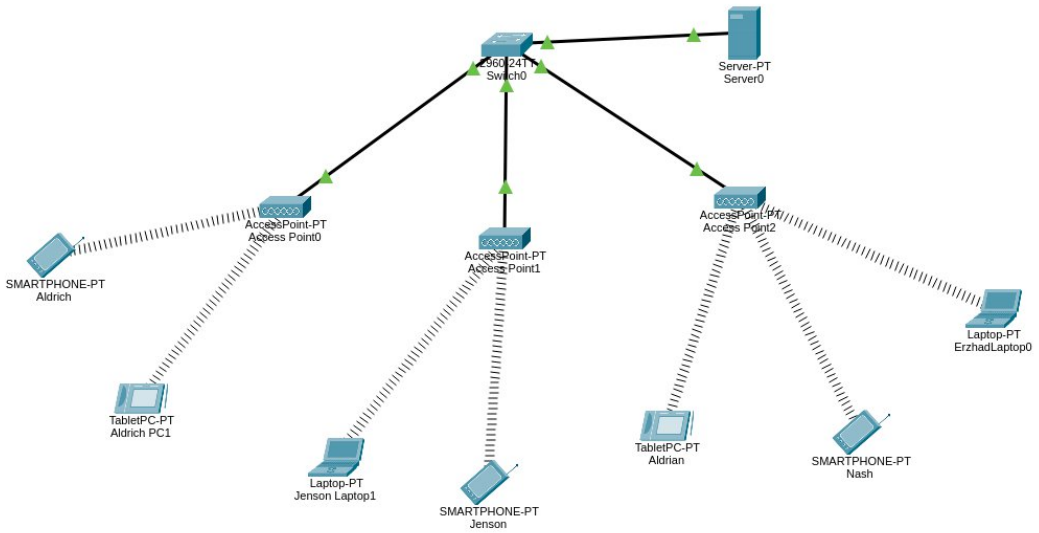
1. Project Overview



This project demonstrates the design and implementation of a **wireless local area network (WLAN)** using Cisco Packet Tracer. The network connects multiple wireless devices through access points, all centralized via a switch, with a server providing network services.

The design ensures **efficient communication, scalability, and centralized management**.The Physical Topology here is a **Star-Bus** or **Tree hybrid**. The Logical Topology is a Multi-Access Broadcast Domain. Even though devices are physically separated by Access Points, they all exist within the same Layer 2 broadcast domain unless you implement VLANs on that switch.

2. Network Topology

Topology Type:

Extended Star Topology (Hybrid Wireless)

Explanation:

- The **switch acts as the central hub**
- Multiple **access points extend wireless connectivity**
- End devices connect **wirelessly through access points**
- A **server is directly connected** to the switch

3. Devices Used

Device Type	Quantity	Purpose
Switch (2960-24TT)	1	Central connection point
Access Points	3	Provide wireless connectivity
Server	1	Provides network services
Smartphones	3	Wireless clients
Laptops	2	Wireless clients
Tablets	2	Wireless clients

4. Network Design Description

The network is structured in a hierarchical manner:

- The **core layer** consists of the **switch**
- The **distribution layer** consists of **3 access points**
- The **access layer** includes all end-user devices

Each access point serves a group of wireless devices:

Access Point 0

- Smartphone (Aldrich)
- Tablet (Aldrich PC1)

Access Point 1

- Laptop (Jenson)
- Smartphone (Jenson)

Access Point 2

- Laptop (Erzhad)
- Smartphone (Nash)
- Tablet (Aldrian)

5. Connectivity

Wired Connections:

- Switch ↔ Server
- Switch ↔ Access Points

Wireless Connections:

- Access Points ↔ End Devices

6. IP Addressing Scheme

Device	IP Address	Subnet Mask
Server	10.10.10.10	255.255.255.0
Clients	DHCP	255.255.255.0

7. Configuration Summary

Switch Configuration:

- Default VLAN used
- Ports connected to access points enabled

Access Point Configuration:

- SSID configured
- Wireless security enabled

End Devices:

- Connected to nearest access point
- Configured via DHCP

8. Advantages of the Design

- ✔ Easy to expand (add more access points/devices)
- ✔ Centralized management via switch
- ✔ Wireless flexibility for users
- ✔ Efficient communication

9. Limitations

- ✖ Single point of failure (switch)
- ✖ Wireless interference possible
- ✖ Limited range per access point

Conclusion

The network successfully demonstrates a **scalable and efficient wireless LAN design**. By integrating access points with a central switch and server, the system ensures reliable communication between multiple wireless devices.